

BERGARUD, Sweden

WMO: 025210

Lat: 58.6003N

Lon: 12.4130E

Elev: 59

StdP: 100.62

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-12.8	-18.1	0.8	-16.4	-14.8	1.0	-12.1	13.0	7.6	11.5	5.5	0.7	320	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	26.2	18.4	24.4	17.4	22.7	16.5	19.5	23.7	18.5	22.4	17.6	21.3	3.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.2	20.9	17.1	12.3	19.9	16.2	11.6	18.9	55.9	24.0	52.6	22.4	49.9	21.4	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.7	7.8	DB	-21.6	29.6	4.8	1.6	-25.1	30.8	-27.8	31.7	-30.5	32.6	-34.0	33.8	(4)
(5)			WB	-21.8	21.3	4.4	0.9	-25.0	22.0	-27.6	22.5	-30.1	23.0	-33.4	23.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-1.6	-1.3	1.4	6.3	10.7	14.4	17.0	16.3	12.2	7.6	3.8	0.2
	DBStd	7.57	5.42	4.77	4.24	3.07	2.96	2.56	2.37	2.35	2.79	3.77	3.99	5.60
	HDD10.0	1682	359	317	267	117	25	1	0	0	12	93	187	304
	HDD18.3	4053	617	551	524	361	236	121	57	70	184	333	436	563
	CDD10.0	696	0	0	1	6	48	132	216	196	78	18	2	0
	CDD18.3	25	0	0	0	0	0	2	14	8	1	0	0	0
	CDH23.3	259	0	0	0	0	6	59	129	64	2	0	0	0
	CDH26.7	29	0	0	0	0	1	7	17	4	0	0	0	0
Wind	WSAvg	3.6	4.3	3.9	3.7	3.7	3.8	3.3	3.0	3.1	3.7	3.4	3.9	3.8
Precipitation	PrecAvg	828	68	55	42	50	50	78	80	83	81	87	79	74
	PrecMax	1136	129	128	98	106	85	126	192	171	214	199	145	156
	PrecMin	619	9	11	7	8	9	27	8	26	20	21	25	13
	PrecStd	127	34	30	22	32	24	32	44	36	50	44	35	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	8.6	14.2	20.3	23.7	27.6	29.0	27.2	23.0	16.9	12.7	10.6
		MCWB	7.0	6.3	8.8	12.5	14.4	18.2	19.7	18.2	15.6	14.0	11.2	9.6
	2%	DB	7.1	7.0	11.0	17.0	21.1	24.9	26.6	25.2	20.4	15.0	11.5	9.1
		MCWB	6.1	5.1	6.4	10.7	14.0	16.8	18.9	18.0	14.4	12.8	10.3	8.2
	5%	DB	5.9	5.7	8.5	14.4	18.8	22.7	24.8	23.3	18.7	13.9	10.6	8.1
		MCWB	4.9	4.5	5.3	9.0	12.7	15.8	18.1	17.1	14.0	12.1	9.5	7.0
	10%	DB	5.0	4.8	6.7	12.3	16.4	20.7	23.0	21.5	17.2	12.8	9.5	7.0
		MCWB	4.1	3.6	4.6	7.9	11.6	15.0	17.3	16.3	13.4	11.4	8.4	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	7.5	8.9	13.1	16.6	19.6	21.2	20.0	16.8	14.7	11.7	9.9
		MCDB	8.1	8.3	13.9	18.1	21.2	25.2	25.9	23.6	19.7	16.0	12.4	10.5
	2%	WB	6.2	5.6	7.1	11.3	14.8	18.0	20.1	18.9	15.7	13.7	10.5	8.5
		MCDB	6.9	6.2	9.8	16.0	18.2	23.2	24.5	22.2	18.3	14.6	11.3	9.1
	5%	WB	5.1	4.7	5.9	9.8	13.7	16.6	19.1	18.1	14.9	12.5	9.7	7.3
		MCDB	5.8	5.5	7.8	13.4	17.5	21.2	23.0	21.6	17.3	13.5	10.5	8.1
	10%	WB	4.1	3.7	5.0	8.5	12.3	15.5	18.1	17.3	14.1	11.7	8.5	5.9
		MCDB	4.9	4.7	6.4	11.5	15.8	19.8	21.8	20.6	16.3	12.8	9.4	6.8
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.4	7.1	9.1	9.0	9.6	9.2	9.2	8.6	6.6	4.7	4.7
		MCDBR	5.2	5.1	9.6	13.4	12.7	13.9	13.2	12.3	11.1	6.8	5.1	4.1
		MCWBR	5.0	4.0	6.2	7.6	6.4	6.7	6.1	5.8	6.0	4.9	4.7	4.2
	5% WB	MCDBR	5.1	4.7	8.2	12.0	10.4	12.2	11.0	9.8	8.9	5.9	4.8	4.0
MCWBR		5.0	3.9	5.7	7.2	5.8	6.6	6.0	5.4	5.7	4.8	4.5	4.1	
Clear-Sky Solar Irradiance	taub	0.281	0.315	0.336	0.374	0.370	0.359	0.380	0.376	0.346	0.334	0.302	0.263	
	taud	2.230	2.306	2.367	2.320	2.398	2.451	2.411	2.420	2.493	2.435	2.301	2.162	
	Ebn at Noon	547	710	809	831	863	878	847	823	796	686	518	431	
	Edh at Noon	46	69	87	107	106	102	104	97	77	62	43	34	
All-Sky Solar Radiation	RadAvg	0.34	0.90	2.43	3.91	5.17	5.84	5.22	4.11	2.78	1.25	0.44	0.23	
	RadStd	0.06	0.16	0.23	0.49	0.44	0.31	0.65	0.36	0.33	0.20	0.09	0.02	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (61 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page